GCSE

Rewarding Learning

Chief Examiner's and Principal Moderator's Report

Digital Technology



Summer Series 2024

Foreword

This booklet outlines the performance of candidates in all aspects of this specification for the Summer 2024 series.

CCEA hopes that the Chief Examiner's and/or Principal Moderator's report(s) will be viewed as a helpful and constructive medium to further support teachers and the learning process.

This booklet forms part of the suite of support materials for the specification. Further materials are available from the specification's microsite on our website at www.ccea.org.uk.

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GCSE DIGITAL TECHNOLOGY

Chief Examiner's Report

Subject Overview

As in previous years, candidates have been well prepared for the examinations and it is clear that there is excellent teaching and learning occurring across the subject area. Across all units, examination performance evidenced good understanding of the key concepts underpinning Digital Technology and the continued development of skills in areas such as multimedia, graphics and programming in a range of languages.

Across the three papers a wide range of learning outcomes were assessed. Questions allowed candidates to respond in a variety of forms. For example, multiple choice, filling blanks, selecting correct responses and constructed/open response questions. Using this wide variety of formats enabled a broad range of candidates to display their knowledge and understanding of the subject throughout the three examination papers. As would be expected, some questions proved to be more discriminating providing stretch and challenge, and many allowed for candidates of all abilities to access marks. This was evidenced in the good range of outcomes at subject level.

There is continued improvement in technical areas such as multimedia design and capabilities in HTML. Those studying the programming route demonstrate improving skills in solution design and the manipulation of complex data structures. All of this is underpinned by the skills gained in Unit 1 where candidates continue to perform extremely well. Across both routes of the qualification, we see computational thinking skills developing and being applied appropriately and in a creative manner. All of this is evidenced in candidate responses across the papers.

Last year we referred to the importance of examination technique. This is still an important aspect of driving future improvement. Where candidates are provided with a context for a question, they should be encouraged to read carefully and attempt to provide answers within that context. At this level contexts are simple and itemised lists or bullet pointed sentences provide clear direction. Candidates should be directed to these lists when reflecting on examination technique.

Another important element which will lead to improved examination performance is the use of good technical language, especially in open response questions. Candidates should be encouraged to make use of subject specific terms when constructing their own responses to questions. Responses which are generic in nature and do not contain the appropriate technical terms will attract fewer marks. This this is particularly true when the Quality of Written Communication is being assessed. Candidates should endeavour to construct responses which make use of the appropriate technical language.

Assessment Unit 1 Digital Technology Compulsory Core

Unit Overview

Overall, the design of the paper was facilitated achievement across a broad range. Assistant examiners saw evidence that it permitted the candidates to achieve good outcomes. The variety and style of the questions catered for all levels of ability. Assistant examiners described the paper as "successful" stating that the "Level of language was clearly suitable for candidates as very few questions were misunderstood or had null responses." Across the paper candidates performed very well and a significant number of candidates were able to access marks in every question on the paper. There was a good range of marks with many falling in the middle and tops bands, demonstrating that candidates were well prepared for the paper. It was good to see candidates attempting the majority of the questions and examiners noted that many of the papers had very few unanswered questions. Overall, this was a well-received paper.

Q1 Multiple Choice

This question was made up of Parts (a) to (j), each part contained four short statements and the candidate was asked to choose the correct response.

(a) Well answered.

(b) Well answered.

(c) Well answered.

(d) Well answered.

(e) Well answered.

(f) Well answered.

(g) Well answered. Some may have found this question a little harder.

(h) Well answered.

(i) Well answered.

(j) Well answered.

Q2 This question was about Cloud Computing generally very well answered by the candidates, most were able to get two out of the four marks. However, the principal examiner would encourage those teaching the topic to encourage the candidates to use more specific technical answers.

Q3 This question was based on Databases and was in the same format that the database question took in 2023. While the easier parts of the question were well attempted it was somewhat disappointing that many candidates in Question 3(b) did not pick up both marks. This part of the question was designed to determine the basic level of knowledge they had about relational databases. Some made very basic errors here. Candidates also seemed to struggle with the basic understanding of a query. While the question was straight forward some candidates struggled with the answer.

Q4 This question was based on Spreadsheets and generally candidates performed well on this question. Marking this question has been tightened and candidates are expected to know formulas better. Again, an area to improve with the candidates in the classroom. Candidates being able to read from a graph is still an area for improvement as this part of Question 4 was poorly attempted.

Q5 This question was on Hardware Components and generally was very straight forward. Question 5(d) saw some candidates lose marks because of wrongly expanding the letter IAS. However, it is essential candidates learn and retain the simple definitions and terms.

Q6 This question was on Networking Topologies, candidates do have a good knowledge of networking, however, some failed to distinguish correctly between the main three tested costing them valuable marks. Part 6(f) was challenging for many candidates. A number were able to name viruses etc. but were unable to add adequate elaboration to obtain further credit.

Q7 This question was on Compression, however, some candidates misunderstood the question and talked about buffering and streaming. Part 7(b) on defining what is 'bit depth', seemed to be another challenging technical term for some candidates. Again, candidates are expected to be able to define/describe the technical terms listed on the specification.

Q8 This question was based on Cybercrime and two areas were specifically targeted, that of Phishing and Denial of Service. The question was styled as an extended writing question and was the question candidates were awarded Quality of Written Communication. Generally, this question was well answered in relation to Phishing, but unfortunately many were unable to describe DOS. This question certainly operated as a question in which to differentiate between abilities.

Q9 This question was about Computer Gaming. This would appear to be a topic enjoyed by most candidates and answers reflected this. However, some did not provide correct answers for Question 9(c). The question asked them to think about hazards other than health. But many gave health answers.

Q10 This question was based on terms used in the specification and their definitions. It was answered very well with many candidates gaining full marks. Candidates seem to enjoy questions where they match words to definitions.

Q11 This question was based on Computer Legislation. The question was very well attempted and many candidates were able to achieve good marks on the question. Having a good mix of tick box type questions and short written responses seems to aid the candidates.

Overall, the paper appeared to be very accessible and the markers were very pleased with the candidate's performance. They also confirmed that the use of language in the questions was at a good level for candidates to access the questions set.

Assessment Unit 2 Digital Authoring Concepts

Unit Overview

In this unit, students enhance their understanding of concepts related to the development and use of digital multimedia systems. This unit accounts for 40% of the GCSE Digital Technology (Multimedia) grade. The exam was worth 120 marks and had a duration of 90 minutes. The questions assessed the skills students developed throughout the unit and covered various digital authoring concepts such as the use of social media and gaming applications, testing and evaluation of digital solutions to problems, the use of HTML, databases, and spreadsheet applications. Overall, the written examination proved accessible to most candidates, with most effectively attempting all questions. More challenging questions effectively differentiated between various levels of understanding.

Candidates showed a proficient level of competence across the paper, managing to secure marks consistently. The majority answered the questions posed and showed a satisfactory grasp of the subject matter. However, an analysis of the responses indicated that some students struggle to provide correct definitions of key terms or to expand acronyms specified in the curriculum.

It should also be noted that open-ended questions included in this written examination are designed to offer candidates the chance to use technical language and gain further marks. These questions provide an opportunity for candidates to display their deeper understanding and ability to articulate complex ideas using the correct terminology. It was, however, observed that many candidates did not fully capitalise on this opportunity. A failure to use correct technical terminology in open-ended questions will limit candidates' ability to gain higher marks in such questions.

Q1 Candidate responses to Part (a) and Part (c) show that despite HTML being a key element of the controlled assessment tasks, evidence shows that many candidates are still unfamiliar with the code constructs needed to develop various list formats and tables. Part (b) and Part (d) were answered well by most candidates.

Q2 Candidates' extensive knowledge of social media applications and prototyping was evident in responses provided in this question.

(a) Most candidates were able to correctly identify two advantages of including interactive features in online gaming applications.

(b) While most candidates correctly identified one disadvantage of including live streaming on a social media application, many failed to gain full marks because they did not describe their example. Instead, they provided two disadvantages, which was not required by the question.

(c) The majority of candidates correctly identified two statements related to a prototype. However, some demonstrated a misconception by incorrectly selecting the statement that referenced the idea that a prototype is a completed version of an application.

Q3 This question was well answered by candidates.

(a) Most candidates correctly named the end user of the proposed website. However, some mistakenly believed the client (the person requesting the new application) to be the end user.

(b) While most students understand the use of design tools in developing multimedia and database applications, a few still confuse timelines and storyboards.

(c) This question was very well answered, with most candidates showing a clear understanding of recording, translating, and executing natural language instructions input via spoken commands.

(d) Common responses included limited command banks, background noise, and the time needed to train the application. Few candidates mentioned technical disadvantages of natural language interfaces. Candidates should ensure distinct examples in such questions; some lost marks by providing similar responses, like training time and user accents.

(e) This question highlighted an issue noted at the start of the report. Some candidates correctly named key GUI features but did not earn maximum marks due to a lack of detail, elaboration, and technical terminology. In some cases, responses did not address the question properly.

Q4 Candidates' understanding of databases in this context has developed over the duration of this specification. There is an increase overall in candidates' understanding of key terms associated with this topic, their ability to develop simple and complex queries using both graphical interfaces and SQL.

(a) All parts of this question were well answered, with most candidates being able to correctly identify a key field, a foreign key and the relationship types used in the database. It should be noted that when referencing relationship types, especially one-to-many, candidates should be encouraged to use the terms listed in the specification as opposed to revering the definition (e.g. many-to-one).

(b) Candidates have shown they have a clear understanding of the terms data integrity and data redundancy. However, when providing a definition of the term data redundancy candidates should be reminded that the term 'repeating unnecessary data' is not the same as 'unnecessary duplication of data'.

(c) Most candidates were able to correctly select the criteria used in the query described in the question stem showing there is an increase in candidate ability to devise complex criteria when querying database tables.

(d) Again, most candidates were correctly able to complete the criteria needed to support the queries described in Part (i) and Part (ii) of this question.

(e) Some candidates are still unable to correctly expand the acronym SQL.

(f) The approach taken to marking the SQL question this year helped increase the accessibility of what has previously been a very difficult topic area. Separating the marks awarded to ensure credit could be given for each individual element of the criteria in the WHERE statement made the question accessible and ensured most candidates were awarded marks in this question. Many candidates were awarded full marks for this difficult and differentiating question.

(g) Candidate responses to this question show there is a clear understanding of the steps involved in the mail merge process. A very well answered question.

(h) While most candidates were correctly able to identify the use of the Report as the tool needed to help users display the results of a query, a small number were unsure of the role of a form when entering data onto a database table.

Q5 **(a)** Responses to Parts (i), (ii) and (iii) of this question show candidates have a clear understanding of the concept of optimisation and its impact upon file size, image quality and upload time.

(b) Candidates remain proficient in the identification of optimised and non-optimised file types as evidenced by responses to this question.

(c) A significant number of candidates were able to demonstrate their understanding of what was meant by the term plugin. Candidates should note the question stem which indicated marks would be awarded for explaining how plugins can help improve cross-platform compatibility. Marks were not awarded for a definition of the term plugin.

(d) Most candidates were able to correctly expand the acronym PDF.

(e) The majority of candidates were able to offer only two pieces of advice to someone developing a cross-platform application.

Q6 **(a)** A well answered question showing candidates have a clear understanding of the type of media one would expect to find in a multimedia application.

(b) While candidate responses showed a general understanding of the meaning of hypertext and scripting, few candidates included enough technical detail in their descriptions to merit full marks in this question.

(c) It was refreshing to see an increased awareness amongst candidates of the various coding constructs applied in script development. Most candidates were awarded full marks for this question.

Q7 **(a)** Most candidates were able to correctly select the correct statement which was true regarding HTML when answering Part (i) of this question.

Part (ii) proved more challenging for candidates. Some candidates focused on the advantages of authoring applications in general (e.g. provision of templates to support the development of professional web sites), as opposed to considering the key advantages of using applications that automatically create HTML for the user.

(b) It was pleasing to see that most candidates were able to correctly match the HTML tags provided with the corresponding tag purpose. In the main, this question was very well answered.

(c) Responses to this question tended to be less website specific and candidates focused more on the use of folders to manage content in a desk top scenario. Candidates should take note of the full question stem and any context provided when developing responses. Candidates were required to link their responses to website development and the management of website content as opposed to general file management when responding to this question.

Q8 Candidates must remember to take note of the context of a question and any additional information provided in the question stem to help them understand question requirements.

(a) Many candidates did not use the information provided in the question stem to correctly identify the type of test data. Proper use of this information is crucial for assessing AO2 in exams. The specification lists valid, invalid, extreme, and null as the four types of test data in a test plan, these terms are expected in responses.

(b) Testing multimedia content is a common question in this unit, and candidates should reference key points in the mark scheme. Some candidates missed full marks in Part (i) and Part (ii) due to insufficient elaboration on the methods for testing images and navigation buttons on a web page.

(c) There was a notable improvement in candidate knowledge and understanding of White Box and Black Box testing. Responses, however, still remain non-technical leaving only a small number of candidates eligible for full marks. Misconceptions regarding which parties are involved in each type of testing, plus lack of elaboration proved to be the main reason for a large proportion of candidates not being awarded four marks in total for Part (i) and Part (ii) combined.

(d) & (e) As with Part (c), candidate increased knowledge and understanding of the A/B and then the Alpha and Beta testing approaches led to many candidates being awarded full marks for these question parts.

Q9 **(a)** It was reassuring to see an increase in candidates' understanding of the role of evaluation in application development and how various documents and individuals support the evaluation process. This question was very well answered.

(b) Candidates remain unsure of the key features of an efficient solution, as evidenced by the non-technical or generic answers provided by many.

Q10 While most candidates were able to correctly match the key words and definitions, a small number remain unsure of the correct definition of event driven in relation to multimedia applications.

Principal Moderator's Report

Controlled Assessment Tasks

All centres successfully submitted controlled assessment tasks via digital submission. Centres **must** ensure candidates have been provided with the correct scenario for Units 3. This changed on an annual basis, released in June of the first year of study. This information is clearly provided on the front cover of the controlled assessment task. Controlled assessment guidelines must be followed throughout tasks and 'modelling' using live data files is not authorised.

Administration

The following procedures **must** be adhered to:

- All candidates of the requested sample, including the **top** and **complete bottom candidate** of the sample must be included.
- Only **essential files** should be submitted for moderation. Original data files or duplicate word processing files that have been converted to **Portable Document Format** should not be included. A small number of centres did not meet this requirement and submitted duplicate files for each candidate.
- There is no requirement to copy and paste the **mark band information** into the eCandidate Record Sheet. Brief, factual comments regarding how marking was allocated is sufficient.

Accuracy of marking

The accuracy of assessment judgements from centres was generally very high and demonstrated a sound understanding of the controlled assessment requirements. A small number of centres marked leniently and marking was adjusted. In some cases, marking was slightly harsh at the lower end of the mark range. Centres should use the 'best fit' mark band approach. The use of templates is prohibited. Candidates are encouraged to produce individual solutions. This is a controlled assessment regulation. Assessment of testing and evaluative sections was an area of concern for some centres.

The majority of centres correctly applied the mark scheme and the banding approach to the assessment of candidate work. It is imperative that all centres complete internal standardisation effectively before submitting marks to CCEA. Some centres did not follow the mark scheme and/or adhere to internal standardisation procedures.

It was positive to note the range of techniques used by candidates in the completion of controlled assessment tasks. There is no requirement for candidates to produce screenshots or step by step reports for controlled assessment tasks. Thank you to all centres for their efforts this series with electronic controlled assessment submission.

Assessment Unit 3 Digital Authoring Practice

Task 1 – Designing the multimedia and database solution using appropriate tools

Many examples of good practice have been noted in the completion of Task 1 by candidates. The following points should be noted:

- The majority of storyboards were computer generated. If 'hand-drawn' storyboards are used, they must be scanned to a high resolution to assist the moderation process. Rather than the focus on the quantity of storyboarding, candidates should focus on the quality and detail of storyboards, for example, the design of the website could include the design of the masterpage. Database design may include a more text/ tabular form/query/report design, to reduce the duplication of work.
- One example of each query, form, report planning is a requirement. More detailed planning on a masterpage would reduce workload, yet still produce a high quality multimedia solution.
- Storyboards must show the sources of information and file types to be used. Some candidates did not provide the level of detail required – image sources, feedback and prototyping missing from more candidates, some plans were missing sufficient detail for third party implementation.
- The design of professional reports is to be commended, with full titles and the company logo applied appropriately.
- Some candidates failed to plan complex queries (two or more criteria) or appropriate reports (grouping, sorting and calculations).
- Some candidates did not attempt prototyping and the use of feedback in refining the solution, this section was marked leniently. 'Marking up' of prototype amendments and feedback could be applied to storyboards in a different font colour.
- Some centres produced unnecessary detail in their design sections e.g. full prototype designs, followed by full storyboards.
- Centres should avoid creating duplicate versions of storyboards to demonstrate prototyping. Centres should encourage candidates to create a prototype and then include annotated screenshots of the final web pages, highlighting amendments from the original design. This approach would streamline the process and clearly showcase the development and refinement stages.

Task 2 – Building the multimedia and database solution

Similarly, many examples of good practice and unique approaches to the task have been noted in the completion of Task 2 by candidates. The following points should be noted:

- If online web editors are used, e.g. Google Sites, the website URL must be presented for moderation with no additional credentials required to access the site, including access to a C2k account. When using Google Sites with a C2k account it is important to ensure that all content is set to shared access including the pages, the form and all other elements housed in the C2k Google Drive.
- Graphics, animation and videos produced for the candidate websites were generally of a high standard.
- Overall, the standard of web authoring was high. It is positive to note many candidates using masterpages and cascading style sheets.
- Scripted elements may include: form validation, slideshow, animation, login, CSS, colour (hex).
- Accessibility elements may include: browser settings, plugins (downloads), speech, magnification, Alt tags.
- Candidates should be encouraged to name their website homepage as 'index' or 'home'.
- Database solutions incorporated appropriate relationships, user friendly and intuitive switchboards, in addition to appropriate validation and input masks.
- Database solutions were implemented from the design document, however, the depth of complex queries and reports (grouping, sorting and calculations) were lacking, particularly grouping and calculations.

Task 3 – Testing the database solution

- This section was marked leniently by some centres.
- Testing must include a range of valid, invalid and extreme data.
- Some candidates designed excessive tests and included screenshots for all testing cases, this could be further refined.
- Corrective action was not completed by a number of centres.
- ***No screenshots are required for testing; screenshots are only required to demonstrate corrective action.***

Task 4 – Evaluating the multimedia and database solution

- This section was marked leniently by some centres.
- Many candidates described their solution, rather than develop evaluative language. A summary of strengths, weaknesses and improvements led to descriptive rather than evaluative comments.
- It is recommended that candidates structure their evaluation logically in terms of the extent to which the user requirements have been met: fully met, partially met or not met.
- Top candidates successfully addressed performance and robustness issues, with refinements/improvements clearly identified. Performance, robustness and refinements were not always given in depth consideration.

All documentation should be submitted as one PDF.

Chief Examiner's Report

Assessment Unit 4 Digital Development Concepts

Unit Overview

This unit accounts for 40% of the GCSE Digital Technology (Programming) grade. The total marks for the exam was 120 marks and the time allowed was 90 minutes. The paper contained nine compulsory questions. As was the case in previous years, a range of questions styles were used in the examination. These included multiple choice, fill the blanks, closed response, tick box, matching terms and open response questions. Marks allocated to parts of questions ranged from one to eight. In Question 6, the candidates' Quality of Written Communication was assessed, this question was worth six marks and was marked within levels; there were three levels describing the features of different types of responses from candidates.

The paper covered a wide range of learning outcomes from the specification for this unit.

Many candidates demonstrated a solid understanding of the key concepts underpinning digital development concepts. In areas such as digital data candidates provided excellent responses when carrying out conversions between the various number bases. They also demonstrated an excellent understanding of the units of data and overflow. A significant proportion of candidates were able to complete the algorithms presented in the examination paper, accurately. Candidates also demonstrated a good understanding of the algorithms presented by selecting the correct answers for output. This in turn shows a solid understanding of the basic concepts related to the construction of algorithms, demonstrating good computational thinking skills.

There was some improvement in responses provided to questions relating to the processing of arrays. Many candidates identified correctly that an array structure contains the data of the same data type and were able to indicate that accessing an array element required the name of the array and the index of the element. A significant majority of candidates now recognise that an array is zero indexed, this was clear from the correct responses when selecting array elements. Whilst candidates demonstrated a good knowledge and understanding of iteration some still find it difficult to correctly implement unbounded iteration through the use of a while loop. Many candidates were unable to set a variable to a value which would control the loop correctly.

Responses indicated that candidates were well prepared for the paper. Questions presented allowed for a broad range of candidates to access marks.

Q1 In this question candidates were presented with an algorithmic solution and were required to predict the outcome given a set of values. This question was generally well attempted with a significant number of candidates achieving the full four marks available.

Q2 Question two was set as a computational thinking exercise. Candidates were asked to consider an algorithm and then to select the correct output given a set of values. This question was less well attempted than question one. The majority of candidates were able to predict the output for Part (a). However, some candidates found difficulty with Part (b) and Part (c). This may have been related to the fact that the loop presented was a WHILE loop and the increment within the loop was two. In general, candidates seem to find the logic associated with the WHILE loop a little more difficult. At GCSE level candidates should be able to use WHILE loops effectively in both the count controlled and condition controlled scenarios.

Q3 This was a scenario-based question.

(a) Candidates were required to complete an algorithm, part of which was presented to them. A high proportion of candidates achieved top marks in this question. The question clearly stated that all IF-statements should contain only one condition. However, some candidates created a solution with IF statements using two or more conditions. This was particularly true in the case of 21–50. The table above the question presented the conditions clearly and the portion of the algorithm provided should have pointed candidates in the direction of the use of the greater than symbol to create single condition statements.

(b) **(i)** Candidates were asked to explain the term validation, this was well attempted with many candidates achieving two marks. In this response candidates were expected to show an understanding that data is checked on input against rules or criteria. Some candidates gave responses like "validation ensures the data is correct", which was not worthy of credit, or "validation means that data meets the criteria of the checks" which was awarded only one mark.

(ii) This part of the question was less well attempted. Some candidates were unable to state clearly how a length check could be used when validating the school name. Most were able to state that the length check would count the number of characters but many did not expand this answer to state that the count could be used to ensure a certain number of characters were not exceeded. Therefore, the candidates had a clear understanding of what a length check was, but many did not expand their answer to include how the length check could be used.

(iii) Candidates were required to select a data type for the Cost per pupil. A high percentage of candidates selected the correct data type. Currency was not an acceptable data type.

(c) Candidates were presented with a portion of an algorithm and asked to complete it. Many candidates achieved top marks in this algorithm. There were areas where some candidates lost marks. In particular, in the IF-statement many candidates lost marks because they were unable to accurately formulate the range check. For example, some candidates used 'groupsize>1' instead "groupsize>=1" for the lower boundary check and similarly left out the '=' in the upper boundary check, losing two marks. It is vital that candidates are familiar with formulating simple range checks using the correct inequalities to ensure the inclusion of the boundary values.

Q4 Overall, this question was extremely well attempted by candidates. A high percentage showing an excellent understanding of converting data between the different number bases.

The overflow question was slightly less well attempted. The question clearly states that candidates should circle overflow. Some candidates did not follow the instructions and therefore lost a mark.

The truth table was well attempted with a significant majority of candidates achieving full marks in this part of the question.

Q5 **(a)** This was a closed response question where candidates had to fill in the blanks from a list provided. Responses indicated that candidates have a solid knowledge of contemporary trends in software development. The majority selected the correct responses related to both procedural and object-oriented programming.

(b) This part of the question related to aspects of software development environments. Whilst many candidates achieved top marks in this part of the question some candidates were unable to correctly identify the features of an interpreter versus a compiler. This is part of high-level code translation and execution. A candidate at this level would be expected to know the difference between the two approaches.

Q6 The Quality of Written Communication was assessed in this question. Candidates were required to explain the difference between execution and logic errors during programme development. In order to achieve marks in the top band, candidates needed to correctly define both execution and logic errors, explain how they affect the programme and give a relevant example of each. There were some excellent explanations which attracted full marks. Candidates falling into the middle band typically were able to explain one of the errors and provide a correct example or explained both of the errors but provided no examples. Candidates in the Level 1 band provided simple explanations or perhaps one valid example but failed to expand adequately on the errors specified.

Q7 This was a scenario-based question, requiring candidates to consider an array structure.

(a) **(i)** This fill in the blanks question was generally well attempted. However, there were still a significant number of candidates who selected 'votes[4]' as the answer to the forth part of the question, indicating that they do not understand that the array is zero indexed. Another common error was the use of 'format' instead of 'data type' for the first answer.

(ii) Well answered with most candidates achieving one mark. Note that 'numeric' was not acceptable.

(b) Candidates evidently found this question on pseudo-code to be slightly more challenging. There were many correct answers achieving two marks. However, a significant number of candidates lost marks for answers that contained reference to code rather than language independent instructions.

(c) **(i)** Candidates were required to initialise the values in the array votes. Use of a loop was not required to achieve this as all of the values were different. Whilst a significant number of candidates achieved the two available marks for any two correct statements, some candidates incorrectly made use of a loop which would not achieve the desired outcome. Simple, correct assignment statements merited full marks on this part of the question.

(ii) This question provided stretch and challenge and candidates achieving in the higher bands across the paper were able to achieve full marks.

Many candidates were unable to express the running total for 'allvotes' or they were unable to correctly specify the loop counter as the index for the array, within the loop. Another common error was placing the number 25 as the upper limit for the FOR loop. Candidates make extensive use of running totals in practice programmes and within their Controlled Assessment. The ability to effectively deploy these statements within solutions is its key skill when completing programmes and as such is expected at GCSE level.

(iii) Again, this was a stretch and challenge question where candidates were required to make use of a WHILE loop. Candidates were provided with contextual information in the form of three bullet pointed sentences. Generally, candidates found it difficult to construct the count controlled WHILE loop. Again, a running total was required within the loop to achieve the required output. This was a more complex task as the running total involved elements from the array. Many candidates lost marks when constructing this statement. A few candidates lost marks by having the output within the WHILE loop – which cost them a mark. However, marks were awarded even if candidates outputted the value of the variable totalvotes.

(iv) In this part of the question candidates were provided with part of an algorithm and three bullet pointed sentences to enable its completion. This question provided stretch and challenge. The bullet pointed sentences provided an explanation of what the algorithm would do. However, many candidates found it difficult to construct the condition within the statement. A significant number were able to correctly express the increment for X within the loop and correctly completed the output statement. Less were able to accurately express the controlling condition (X < 5/X < = 4). This emphasises again that there is an area for development in the use of unbounded iteration.

(d) **(i)** Well attempted by most candidates. However, a significant number of candidates selected Option 2 “The bubble sort takes each element and places it in the correct place in a sorted sub-list” as true.

(ii) Many candidates achieved four marks in this question which required them to simulate a number of passes of the bubble sort. Candidates achieving in the top and middle bands were able to do this. Even the weaker candidates achieved one mark for completion of PASS 4 showing the fully sorted array.

(e) **(i)** Candidates were able to explain clearly what a linear search is.

(ii) Candidates made a good attempt at answering this question about the binary search. Acceptable responses included reference to the following: the the binary search requires a sorted list, finds the midpoint of the data, repeatedly reduces the search list by half, compares the midpoint to the target value or the search ends when a match is found. Candidates making two of these points achieved full marks.

(iii) This one mark question required candidates to state why Joan’s data was more suited to a binary search. Less candidates than expected did not recognise that the data having been sorted made it more suitable for using with the binary search.

Q8 **(a)** Candidates were again required to identify suitable data types for use in a solution. Sample data was provided, and the majority of candidates recognised that the data types required were string and Boolean. Note that Y/N was not acceptable as a data type.

(b) **(i)** Some candidates did not recognise this type of testing as unit testing. Quite a number identified it as integration testing.

(ii) This question provided candidates with part of a test plan which had to be completed. Generally speaking, this question was very well attempted by a wide range of candidates. The significant majority were able to correctly identify the reason for tests, provide suitable correct test data and predict the expected outcome. This is clearly an area of strength for candidates which is supported by the controlled assessment element.

(c) Candidates found this question quite challenging. Whilst many correctly identified who should carry out both black-box and white-box testing, answers on the right-hand side of the table were less well articulated. Some candidates correctly stated that black-box testing tests the 'inputs and outputs' of the system and white-box testing 'tests the internal structure'. However, they struggled to adequately expand on either of these answers to achieve the second mark.

Q9 **(a)** This closed response question on evaluation was very well attempted by the broad range of candidates. Many achieved full marks. Common errors included inserting 'solution' instead of 'user requirements' or inserting 'user interface' instead of 'solution'.

(b) Candidates were required to identify how a system's robustness can be evaluated. Whilst this was well answered and many candidates referenced mark scheme answers, some candidates incorrectly referenced types of testing e.g. black-box testing or beta testing.

Principal Moderator's Report

Controlled Assessment Tasks

All centres successfully submitted controlled assessment tasks via digital submission. Centres **must** ensure candidates have been provided with the correct scenario for Unit 5. This changed on an annual basis, released in June of the first year of study. This information is clearly provided on the front cover of the controlled assessment task. Controlled assessment guidelines must be followed throughout tasks and 'modelling' using live data files is not authorised.

Administration

The following procedures **must** be adhered to:

- All candidates of the requested sample, including the **top** and c**omplete bottom candidate** of the sample must be included.
- Only **essential files** should be submitted for moderation. Original data files that have been converted to **Portable Document Format** should not be included. A very small number of centres did not meet this requirement.

Accuracy of marking

The accuracy of assessment judgements from centres was sound and demonstrated a sound understanding of the controlled assessment requirements. High quality work was produced by centres and application of the mark scheme was highly accurate. No centres were adjusted this series.

The use of templates is prohibited. Candidates are encouraged to produce individual solutions. Assessment of testing and evaluative sections continue to be an area of concern for some centres, regarding leniency.

The majority of centres correctly applied the mark scheme and the banding approach to the assessment of candidate work. It is imperative that all centres complete internal standardisation effectively before submitting marks to CCEA. Some centres did not follow the mark scheme and/or adhere to internal standardisation procedures.

It was positive to note the range of techniques used by candidates in the completion of controlled assessment tasks. There is no requirement for candidates to produce screenshot reports for controlled assessment tasks. Thank you to all centres for their efforts this series with electronic controlled assessment submission.

Assessment Unit 5 Digital Development Practice

Task 1 – Design a solution using appropriate tools

Many examples of good practice have been noted in the completion of Task 1 by candidates. The following points should be noted:

- In some instances, user requirements were extracted from the task, without development or interpretation. Target audience needs must also be fully identified.
- The design component was particularly well implemented with high quality flowcharts and pseudo code included effectively.
- Validation and error trapping techniques were designed appropriately, however, use of user feedback to refine the solution was omitted by a number of centres and marked leniently.

Task 2 – Building a solution

Similarly, many examples of good practice and unique approaches to the task have been noted in the completion of Task 2 by candidates. Centres are to be highly commended upon their approach to this task. The following points should be noted:

- **Program usernames and passwords must be noted on the eCandidate record sheets. A number of centres are not adhering to this essential element of controlled assessment submission.**
- Marking was highly accurate across the sampled centres and the programming solutions were of a high standard.
- Candidates should be encouraged to develop their own program style and content.
- Solutions of the highest standard gave due consideration to system robustness.
- All solutions must be annotated.

Task 3 – Testing the solution

- Test plans were well structured, in tabular format (showing tests, expected outcome, actual outcome and fixes), in some cases test plans were cluttered and could be more professionally presented.
- It is very important that testing incorporates a range of tests (black and white box testing) and includes valid, invalid and extreme test data. Extreme testing of data was lacking in a number of centres.
- Some candidates designed excessive tests and included screenshots for all testing cases.
- No screenshots are required for testing; screenshots are required to demonstrate corrective action. Corrective action was not completed by a number of centres.

Task 4 – Evaluating the solution

- Some candidates described their solution, rather than develop evaluative language. A summary of strengths, weaknesses and improvements led to descriptive rather than evaluative comments.
- It is recommended that candidates structure their evaluation logically in terms of the extent to which the user requirements have been met: fully met, partially met or not met.
- Top candidates successfully addressed performance and robustness issues, with refinements/improvements clearly identified. Performance, robustness and refinements were not always given in depth consideration.
- A small number of centres marked performance, robustness and refinements section leniently.

All documentation should be submitted as one PDF.

Contact details

The following information provides contact details for key staff members:

- **Specification Support Officer: Nuala Tierney**
  (telephone: (028) 9590 6689, email: ntierney@ccea.org.uk)
- **Officer with Subject Responsibility: Andrew Douglas**
  (telephone: (028) 9590 6701, email: adouglas@ccea.org.uk)

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COUNCIL FOR THE CURRICULUM, EXAMINATIONS AND ASSESSMENT
29 Clarendon Road, Clarendon Dock, Belfast BT1 3BG
Tel: +44(0)28 9026 1200 Email: info@ccea.org.uk
Web: ccea.org.uk

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